



Cambridge IGCSE™

MATHEMATICS

0580/42

Paper 4 (Extended)

May/June 2022

MARK SCHEME

Maximum Mark: 130

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **9** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfw	not from wrong working
soi	seen or implied

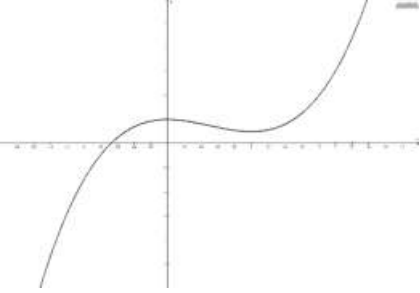
Question	Answer	Marks	Partial Marks
1(a)	150	2	B1 for answer $150k$ or M1 for prime factors of 30 or 75 seen or a list of multiples of both 30 and 75 with at least 3 of each or for $\frac{30 \times 75}{15}$ oe or for answer $2 \times 3 \times 5^2$
1(b)	152 190 266	3	Accept in any order B2 for two correct answers or M1 for $\frac{608}{4+5+7} \times k$ oe where $k=1, 4, 5, 7$
1(c)	2.61×10^{-2} 2.61×10^{-2} or $2.608... \times 10^{-2}$	2	B1 for figs 2608 or 261 seen If 0 scored, SC1 for answer $2.6[0] \times 10^{-2}$ without more accurate value in standard form seen
1(d)	$\frac{27}{99}$ oe fraction	1	
1(e)	2.8	1	
	g/cm^3 or g cm^{-3}	1	
2(a)	$PQR = 90$ angle in semi-circle	B1	
	$PRQ = 61$ angle sum of triangle [= 180]	B1	
	$PSQ = 61$ angle in same segment	B1	If 0 scored SC1 for $PSQ = PRQ$ [= 61] soi
2(b)	57	4	B1 for $ABT = 98$ B1 for TAB or $ATB = 41$ B1 for $BTC = 41$ or $TBC = 82$ or $ATC = 82$ soi
3(a)	8.25 or 8.246...	3	M2 for $(3 - -5)^2 + (2 - 4)^2$ oe or better or M1 for $(3 - -5)$ and $(2 - 4)$ oe seen
3(b)	$[y =] 4x + 7$	5	B1 for [midpoint] $(-1, 3)$ soi M1 for [gradient of $l =$] $\frac{4-2}{-5-3}$ oe M1 for gradient -1 / <i>their</i> $\left(-\frac{1}{4}\right)$ M1dep on at least M1 for <i>their</i> $(-1, 3)$ substituted into $y = \text{their } m \times x + c$ oe

Question	Answer	Marks	Partial Marks
3(c)	$(0, -8)$ and $(0, 16)$	4	B3 for $(0, -8)$ or $(0, 16)$ or for -8 and 16 OR B2 for distance = $[±]12$ soi or M1 for $13^2 - (5[-0])^2$ oe B1 for both answers $(0, k)$, $k \neq 0$ or 4 ALT METHOD B3 for $(0, -8)$ or $(0, 16)$ or for -8 and 16 OR M2 for $y^2 - 8y - 128 [= 0]$ or for $(y - 4)^2 = 144$ or better or M1 for $13^2 = (-5 - 0)^2 + (4 - y)^2$ oe B1 for both answers $(0, k)$, $k \neq 0$ or 4
4(a)	7.06 or 7.058... or 7.059	3	M2 for $\sqrt{6.4^2 + 10.9^2 - 2 \times 6.4 \times 10.9 \times \cos 38}$ oe OR M1 for $6.4^2 + 10.9^2 - 2 \times 6.4 \times 10.9 \times \cos 38$ oe A1 = 49.8...
4(b)(i)	97	1	
4(b)(ii)	15.3[0...]	3	M2 for $[AB =] \frac{10.9 \times \sin their\ 97}{\sin 45}$ or M1 for $\frac{\sin their\ 97}{AB} = \frac{\sin 45}{10.9}$ oe
4(c)	72.8 to 72.81...	3	M2 for $\frac{1}{2}(6.4) \times 10.9 \times \sin 38 + \frac{1}{2} their\ 15.3 \times 10.9 \times \sin 38$ oe or M1 for $\frac{1}{2} \times 6.4 \times 10.9 \times \sin 38$ oe or $\frac{1}{2} \times their\ 15.3 \times 10.9 \times \sin 38$ oe or M1 for height = $10.9 \times \sin 38$ oe
5(a)	Correct lines drawn	2	B1 for one correct with no incorrect lines
5(b)(i)(a)	Translation or translate $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$ oe	2	B1 for each
5(b)(i)(b)	Rotation or rotate 90 [anticlockwise] oe [centre] (2, 1)	3	B1 for each
5(b)(ii)(a)	Triangle at $(-5, 6)$ $(-2, 6)$ $(-2, 5)$	2	B1 for reflection in $y = k$

Question	Answer	Marks	Partial Marks
5(b)(ii)(b)	Triangle at (1, 5) (1, 7) (7, 7)	2	B1 for correct size and orientation, wrong position
6(a)	42 028	2	M1 for $\frac{380}{500}$ oe soi isw
6(b)	$\frac{47}{66}$ oe	4	0.712[1...] M3 for $2\left(\frac{5}{12} \times \frac{4}{11}\right) + 2\left(\frac{4}{12} \times \frac{3}{11}\right) + 2\left(\frac{5}{12} \times \frac{3}{11}\right)$ oe or $1 - \left(\frac{5}{12} \times \frac{4}{11} + \frac{4}{12} \times \frac{3}{11} + \frac{3}{12} \times \frac{2}{11}\right)$ oe or M2 for sum of 3 or more correct product pairs and no incorrect pairs or for $\frac{5}{12} \times \frac{4}{11} + \frac{4}{12} \times \frac{3}{11} + \frac{3}{12} \times \frac{2}{11}$ and no other pairs or M1 for $\frac{k}{12} \times \frac{j}{11}$ seen If 0 scored SC1 for answer $\frac{94}{144}$ oe
6(c)	52	2	M1 for $x \times \frac{100-16}{100} = 43.68$ oe or better
6(d)(i)	70 or 70.16[5...] or 70.17 or 70.2	3	M2 for $\frac{29750 \text{ to } 29800}{400+25}$ or $\frac{29750 \text{ to } 29800}{400+24}$ or $\frac{29800-50}{400 \text{ to } 425}$ or B1 for 29 750 or 29 850 or 29 849 or 375 or 425 or 424 seen
6(d)(ii)	2399 or 2400 nfw	2	B1 for 27 450 or 27 550 or 27 549 or 29 850 or 29 849 seen
7(a)	25.2 or 25.23...	4	M1 for midpoints soi M1 for use of $\sum fx$ with x in correct interval including both boundaries M1 (dep on 2 nd M1) for $\sum fx \div 150$
7(b)	5 correct blocks	4	B3 for 4 correct blocks or B2 for 3 correct blocks or B1 for 2 correct blocks or block widths 10, 10, 5, 15, 10 If 0 scored SC1 for 4 correct frequency densities from 1.2, 3.8, 6.4, 3.33[3...] and 1.8 oe soi
7(c)(i)	12, 50, 82, 132, 150	2	B1 for 3 or 4 correct

Question	Answer	Marks	Partial Marks
7(c)(ii)	92	2	M1 for $150 - 12$ oe seen If 0 scored, SC1 for answer 8[%]
8(a)	$\frac{1}{2}$ or 0.5 oe	2	M1 for $10 - 3 = 11p + 3p$ oe or better
8(b)	$[m =] \frac{2k}{c^2 - g}$ oe final answer	3	M1 for correctly isolating m terms M1 for correctly factorising M1 for dividing by a bracket with two terms to the final answer Maximum mark M2 if final answer incorrect
8(c)	0 4.5 oe	5	B4 for $2x^2 - 9x [= 0]$ or $9x - 2x^2 [= 0]$ or better OR M2 for $(2x + 3) + 4(x - 3) = (x - 3)(2x + 3)$ or better or M1 for $(2x + 3) + 4(x - 3)$ seen oe or common denominator $(x - 3)(2x + 3)$ oe B1 for $2x^2 - 6x + 3x - 9$ or better seen
8(d)	$y^2 - 10y + 21 [= 0]$ or $x^2 - 4x - 12 [= 0]$	M2	M1 for $y^2 + 5(12 - 2y) = 39$ oe or $5x + \frac{(12 - x)^2}{2^2} = 39$ seen oe
	$(y - 3)(y - 7) [= 0]$ or $(x + 2)(x - 6) [= 0]$	M1	or for correct factors for <i>their</i> 3– term quadratic equation or for correct substitution into quadratic formula or correctly completing the square for <i>their</i> 3– term quadratic equation
	$x = -2$ $y = 7$ $x = 6$ $y = 3$	B2	B1 for $x = -2, x = 6$ or for $y = 7, y = 3$ or for one correct pair of x and y values
8(e)	$2x^3 + x^2 - 54x + 72$ final answer	3	B2 correct expansion of three brackets unsimplified or for final answer of correct form with 3 out of 4 terms correct or B1 correct expansion of two brackets with at least three terms out of four correct
9(a)	$PMR = MSR = \text{right angle[s]}$ or 90°	B1	
	$PRM = MRS$ same angle	B1	
	AAA oe OR $MPR = SMR$ 3rd angle of triangle	B1	Dep on B1B1 and no errors seen

Question	Answer	Marks	Partial Marks
9(b)(i)	5.5	2	M1 for $\frac{x}{4.5} = \frac{9.9}{8.1}$ oe
9(b)(ii)	16.7 or 16.73 to 16.74	2	M1 for $25 \times \left(\frac{8.1}{9.9}\right)^2$ oe or $25 \times \left(\frac{4.5}{\text{their } 5.5}\right)^2$ oe
10(a)	1, 2, 3	2	M1 for $15 - 8 > 5n - 3n$ oe If 0 scored, B1 for 2 correct answers and no others or 3 correct answers with one extra value
10(b)(i)	$10y + 8x \leq 80$ oe final answer $x > 4$ oe final answer $2y > x - 4$ oe final answer	3	B1 for each If 0 scored, SC1 for $10y + 8x < 80$ oe final answer and $x \geq 4$ oe final answer and $2y \geq x - 4$ oe final answer
10(b)(ii)	23 final answer	2	M1 for 7 and 2 selected soi
11(a)(i)	4.455 to 4.456... [= 4.46]	2	M1 for $[r =] \frac{28}{2\pi}$ oe
11(a)(ii)	1250 or 1247 to 1249.9...	2	M1 for $20 \times \pi \times 4.46^2$ oe
11(a)(iii)	66[.0] or 65.95 to 66.02	3	M2 for $[\tan] = \frac{20}{2 \times 4.46}$ oe or B1 for identifying angle ANB on cylinder not on rectangle
11(b)	11.8 or 11.82 to 11.83	5	M2 for $[r =] \sqrt[3]{\frac{310 \times 3}{2\pi}}$ oe or $[h =] \sqrt[3]{\frac{310 \times 3 \times 4}{\pi}}$ oe or M1 for $310 = \frac{1}{3}\pi \times r^2 \times 2r$ or $310 = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h$ M2 for $\sqrt{(\text{their } r)^2 + (2 \times \text{their } r)^2}$ oe or M1 for $[l^2 =] (\text{their } r)^2 + (2 \times \text{their } r)^2$ oe

Question	Answer	Marks	Partial Marks
12(a)	$3x^2 - 2kx$	M2	M1 for $3x^2$ or $-2kx$
	<i>their</i> $\frac{dy}{dx} = 6$	M1	Dep on at least M1 for derivative
	$x = 2$ substituted in <i>their</i> $\frac{dy}{dx}$	M1	Dep on at least M1 for derivative
	Correct working leading to 1.5 oe	A1	A0 if any errors in working leading to 1.5
12(b)	(0, 1) (1, 0.5)	4	B3 for $x = 0$ and $x = 1$ or for (1, 0.5) OR M1 for <i>their</i> $\frac{dy}{dx} = 0$ B1 for $3x^2 - 3x$ oe or better
12(c)	correct sketch 	2	with max on positive y-axis and min in 1st quadrant B1 for positive cubic or for graph with one max which is on pos y-axis and one min which is in 1st quadrant